Modeling the Relationship Between Daily Activity Metrics and Apple Health Active Energy

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***Abstract*—This study models the within-person day-level relationship between wearable-derived daily summaries and Apple Health Active Energy (kcal). Daily data were collected over 71 consecutive days (2025-11-24 to 2026-02-02) and curated into two daily datasets: Gold Core (12 columns) and Gold Extended (26 columns) using a strict daily spine and quality gates. Suspicious zero-energy days were treated as likely non-wear and converted to missing values (11 days): a day was flagged when both Active Energy and Resting Energy were exactly 0, and related activity and heart-rate summaries were also blanked as likely invalid. Due to missingness across predictors and the target, supervised modeling used a complete-case subset of 46 days (25 rows dropped), split chronologically into 36 training days (2025-11-25 to 2026-01-21) and 10 test days (2026-01-22 to 2026-02-02). The modeling design matrix used 19 predictors: same-day summaries, three lag-1 features, and day-of-week indicators. Exploratory analysis showed strong positive relationships between Active Energy and movement-derived variables (e.g., stand hours, distance, steps, and exercise minutes). Two baselines (dummy mean and naive "yesterday") and two regressors (OLS Linear Regression and Random Forest with 500 trees, random_state = 42) were evaluated using MAE and R^2, with walk-forward time-series cross-validation (TimeSeriesSplit, 5 folds). On the held-out test set, Random Forest achieved MAE = 29.19 kcal and R^2 = 0.9448, outperforming Linear Regression (MAE = 50.37 kcal, R^2 = 0.8619). Walk-forward cross-validation yielded lower and more variable performance for Random Forest (R^2 mean = 0.8396, SD = 0.1810), consistent with small-sample instability in longitudinal personal data. Results indicate that Apple Health daily summaries explain Apple Health's same-day Active Energy estimate well in this single-subject setting, with key limitations including missingness/non-wear, temporal dependence, and construct overlap between predictors and the target estimate.**

***Index Terms*—Apple Health, Apple Watch, active energy, personal informatics, time series, regression, Random Forest, missing data**

I. Introduction

The aim of this study is to examine how daily wearable-derived activity and physiology summaries (e.g., steps, distance, exercise minutes, stand hours, and heart-rate statistics) relate to Apple Health Active Energy (kcal) at the day level. The purpose is to determine whether the daily summary metrics people commonly see in Apple Health can be used to explain and predict the day-to-day changes in Active Energy within a single individual.

This paper evaluates both association and prediction by combining exploratory relationship analysis (e.g., correlation) with supervised regression modeling for same-day explanatory prediction. For clarity and consistency, this paper refers to the metric as Apple Health Active Energy (kcal).

Real-world wearable datasets are often messy. People forget to wear their watch, devices run out of battery, workouts may be logged inconsistently, and some signals (such as HRV or respiratory rate) may appear only on certain days depending on sleep, sensor quality, or settings. Even when data exists, exports from different sources (e.g., Apple Health XML versus third-party export tools) can disagree or duplicate records, creating inflated or unreliable totals. These issues complicate modeling because missingness and measurement quirks can bias results toward "good recording days."

In this study, the participant did not add manual entries; all metrics were passively recorded by Apple Watch/iPhone.

The primary objective of this study is to predict Apple Health Active Energy (kcal) from daily activity summaries using same-day explanatory models with time-aware evaluation and baseline comparisons. This supports the objectives to: (1) construct reproducible daily datasets from Apple Health exports using transparent data-quality checks; (2) identify which daily activity variables are most strongly associated with Active Energy (kcal) using exploratory analysis; and (3) assess how missing data and time order affect evaluation and interpretation. This is guided by one primary research question: How accurately can Active Energy (kcal) be predicted from daily activity?

This study is needed because Apple Health Active Energy is itself an algorithmic estimate derived from wearable signals, yet users often interpret it as a direct "outcome." By building a transparent daily dataset with explicit quality checks and by evaluating time-aware prediction models against simple baselines, this work clarifies what can (and cannot) be inferred from Apple Health daily summaries.

This work uses a single-participant, researcher-as-subject (self-tracking) design, where the researcher is also the participant (n = 1). Accordingly, results characterize within-person relationships over time rather than population-level generalization.

II. Literature Review

This study focuses on predicting Apple Health Active Energy (kcal) using daily Apple Watch and Apple Health summary metrics (such as steps, distance, exercise minutes, stand hours,

and heart-rate features). Because Active Energy is an estimate produced from wearable sensor data and user information, it is important to review prior work that explains (1) how Apple Watch collects activity and heart data and converts it into calorie estimates, and (2) how accurate these outputs are in real-world, free-living conditions. The related literature below highlights Apple Watch sensing and algorithm approaches, validation findings (e.g., for physical activity and activity energy expenditure), and common real-life data issues that can affect daily records—providing the foundation for this study's objectives and analysis.

Kwon et al. (2021) noted that Apple Watch is widely used and relies on onboard sensors (e.g., accelerometer, gyroscope, altimeter) to estimate daily activity outcomes such as steps, distance, exercise time, and activity energy expenditure. They also described how the Activity app summarizes daily movement behaviors (move, exercise, stand), which aligns with the daily metrics used as predictors in this study. However, Kwon et al. emphasized that the accuracy of Apple Watch estimates in free-living conditions may vary, supporting the need for analyses that examine how these daily summaries relate to Apple Health Active Energy in real-world data [1].

Apple Inc. (2024) described how Apple Watch estimates heart rate and calorie-related outputs using multiple data sources, including motion sensors (e.g., accelerometer/gyroscope), heart rate signals, workout context, and user characteristics (e.g., age, sex, height, weight). The report also discusses factors that can reduce measurement quality in everyday use, such as poor watch fit, motion artifacts, environmental conditions, or incomplete wear time. This is directly relevant to the current study because it explains why daily exports may contain missing or unreliable values and why the same-day activity summaries used as predictors should be strongly related to Active Energy, which is itself derived from similar underlying signals [2].

A systematic review and meta-analysis by Choe and Kang (2025) synthesized evidence across studies evaluating Apple Watch performance for metrics including heart rate and energy expenditure. Overall, the review suggests Apple Watch can provide useful estimates for monitoring and research applications, while accuracy depends on the outcome, activity type, and testing conditions. These findings support the use of Apple Watch-derived measures in this study, while reinforcing that Active Energy should be interpreted as an estimate influenced by context and data quality in free-living settings [3].

This work does not validate energy expenditure against gold-standard calorimetry; it models Apple Health's Active Energy estimate using other Apple Health summaries.

III. Methodology

This section describes how the study data was collected, curated, and analyzed in a reproducible way. The workflow converts raw Apple Health exports into daily datasets (Gold Core and Gold Extended), performs exploratory analysis and descriptive statistical tests, and trains same-day explanatory regression models to predict Apple Health Active Energy (kcal) from daily summary metrics.

A. Participant and Data Collection

This study employed a single-subject, researcher-as-subject design (n = 1). The participant is a 23-year-old fourth-year student currently pursuing a Bachelor of Science in Computer Science at National University–Manila and resides in Metro Manila. Data were collected passively using an Apple Watch paired with an iPhone through the Apple Health framework. The study period covered 2025-11-24 to 2026-02-02 (71 consecutive days), with all variables aggregated at the day level (one row per day).

Data sources consisted of Apple Health *export.xml* (record-level samples aggregated into daily features) and Health AutoExport CSV files (daily summaries and selected measures). Steps and distance were taken from AutoExport as the canonical source. For shared daily summary columns (energy, exercise, stand, flights, and daily heart-rate summary statistics), the pipeline preferred AutoExport with XML fallback when an AutoExport value was missing for a given day. Heart-rate sample count was derived from XML aggregation as the sole source. Additional physiology variables (HRV, respiratory rate, resting heart rate, walking speed, step length, and stand minutes) were collected from AutoExport when available. No manual entries were added by the participant; metrics were passively recorded by Apple Watch/iPhone.

B. Operational Definitions

This subsection defines each variable as used in the curated daily datasets. All variables are aggregated at the **day level**.

1) Gold Core (primary analysis dataset): Gold Core is the primary dataset for EDA and modeling and contains 12 columns (including date). Table I lists the Gold Core variables.

TABLE I: Gold Core Variables (Daily Aggregation)

Variable Name	Unit	Description
Steps	count/day	Total daily step count.
Distance	km/day	Total daily walking + running distance.
Active Energy	kcal/day	Active energy burned; modeling target.
Resting Energy	kcal/day	Resting energy burned.
Exercise Minutes	min/day	Minutes credited as exercise.
Stand Hours	hours/day	Hours with at least 1 minute of standing.
Flights Climbed	count/day	Floors climbed.
Mean Heart Rate	bpm	Mean heart rate for the day.
Min Heart Rate	bpm	Minimum heart rate for the day.
Max Heart Rate	bpm	Maximum heart rate for the day.
Heart Rate Samples	count/day	Number of heart-rate samples recorded that day.

Note: Data extracted from Apple Health exports.

2) Gold Extended (secondary dataset for data-quality inspection): Gold Extended includes Gold Core plus additional physiology summaries and contains 26 columns (including date). These extended variables were used for missingness inspection and descriptive exploration, but were excluded from

the primary modeling feature set due to high missingness. Table II lists the additional variables.

TABLE II: Gold Extended Additional Physiology Variables

Data extracted from Apple Health exports.

Variable Name	Unit	Description
HRV	ms	Heart-rate variability (SDNN).
Respiratory Rate	breaths/min	Respiratory rate.
Resting Heart Rate	bpm	Resting heart rate.
Walking Speed	km/h	Average walking speed.
Step Length	cm	Average walking step length.
Stand Minutes	min/day	Total standing minutes.

C. Data Cleaning and Preprocessing

Cleaning and preprocessing were performed prior to statistical tests or modeling to enforce a consistent daily structure and reduce artifacts from incomplete wear time.

A strict daily spine was created for all calendar days from 2025-11-24 to 2026-02-02. If a given source contained no record for a day, that day was retained with missing values rather than removed.

Suspicious zero-energy days were treated as likely non-wear or failed recording. Specifically, any day in which both Active Energy and Resting Energy were exactly zero was converted from 0 to missing (NaN). On flagged days, related summaries (Exercise Minutes, Stand Hours, and daily heart-rate aggregates: mean/min/max) were also blanked as likely invalid. This occurred on 11 days.

Steps and distance from Health AutoExport were treated as the canonical source because Apple Health XML exports can contain overlapping records that inflate daily totals. XML-derived step or distance totals (if computed) were retained only for auditing and excluded from analysis.

A final plausibility check was conducted by computing a meters-per-step ratio to screen for obvious aggregation errors. Values were inspected against a typical human range of approximately 0.3–1.5 m/step; no rows were removed based on this check (meters-per-step mean = 0.7127; range 0.574–0.829; outliers outside 0.3–1.5: 0).

Supervised modeling used a complete-case subset because the selected models require non-missing values. After feature engineering (lag features and day-of-week encoding), listwise deletion (dropna) yielded 46 usable days for modeling (25 rows dropped from the 71-day spine).

D. Exploratory Data Analysis

EDA characterized temporal patterns, distributions, pairwise relationships, multicollinearity among candidate predictors, and data availability. The workflow included time-series plots for core metrics, histograms and boxplots for distribution inspection, a steps-versus-active-energy scatter plot, a Pearson-correlation heatmap across Gold Core variables, and missingness summaries for Gold Core and Gold Extended.

E. Statistical Analysis and Modeling

The analysis combined descriptive visualization, correlation-based relationship assessment, and supervised regression modeling with time-aware evaluation.

Pearson correlation was computed between steps and active energy on rows where both variables were non-missing. Because normality was not guaranteed, Shapiro–Wilk tests were computed for steps and active energy, and Spearman correlation was included as a rank-based robustness check. Welch's t-test compared weekday versus weekend active energy under unequal-variance assumptions. Because observations are daily measurements from a single individual, serial dependence is expected; therefore, p-values are interpreted as descriptive.

The modeling objective was *same-day explanatory prediction* of Active Energy from daily summaries. Feature engineering included lag-1 variables (steps_lag1, exercise_min_lag1, active_energy_kcal_lag1) and day-of-week one-hot encoding (dow_1 through dow_6, with one day omitted as the baseline category). The complete-case modeling design matrix used 19 predictors (exact order): *steps, distance_km, resting_energy_kcal, exercise_min, stand_hours, flights_climbed, hr_mean_bpm, hr_min_bpm, hr_max_bpm, hr_samples, steps_lag1, exercise_min_lag1, active_energy_kcal_lag1, dow_1, dow_2, dow_3, dow_4, dow_5, dow_6*.

Data were split chronologically using an 80/20 train/test division to avoid future leakage: train N = 36 (2025-11-25 to 2026-01-21) and test N = 10 (2026-01-22 to 2026-02-02).

Four predictors were evaluated: a dummy mean baseline; a naive yesterday baseline defined as:

$$\hat{y}_t = y_{t-1};$$

an OLS Linear Regression model; and a Random Forest Regressor with n_estimators = 500 and random_state = 42 (other settings default). Performance was assessed using Mean Absolute Error (MAE) and R^2. In addition to held-out test evaluation, walk-forward time-series cross-validation using TimeSeriesSplit was performed with 5 folds (n_splits = 5; no gap; no fixed test_size). Fold sizes (train →test) were 11→7, 18→7, 25→7, 32→7, and 39→7. The naive yesterday baseline was omitted from CV due to fold-boundary alignment requirements for lagged target values.

Results are interpreted with three measurement considerations: missingness/non-wear can introduce selection bias; within-person daily data are temporally dependent; and Active Energy is an Apple Health algorithmic estimate derived from overlapping movement-related signals, so strong same-day prediction reflects internal consistency rather than ground-truth validation.

IV. Results

This section reports patterns from EDA, descriptive statistical tests, missingness inspection, and model evaluation on the chronological held-out test set.

A. Exploratory Patterns and Distributions

Fig. 1 shows day-to-day trajectories for core activity and energy variables. Steps, distance, and active energy vary substantially across days with roughly synchronized peaks and troughs, consistent with shared underlying movement intensity.

Fig. 2 and Fig. 3 summarize distributions of core metrics using histograms and boxplots. Flights climbed is right-skewed with many near-zero days, while resting energy shows comparatively lower variance than movement-derived metrics. No outliers were removed; downstream exclusions occurred only through complete-case filtering.

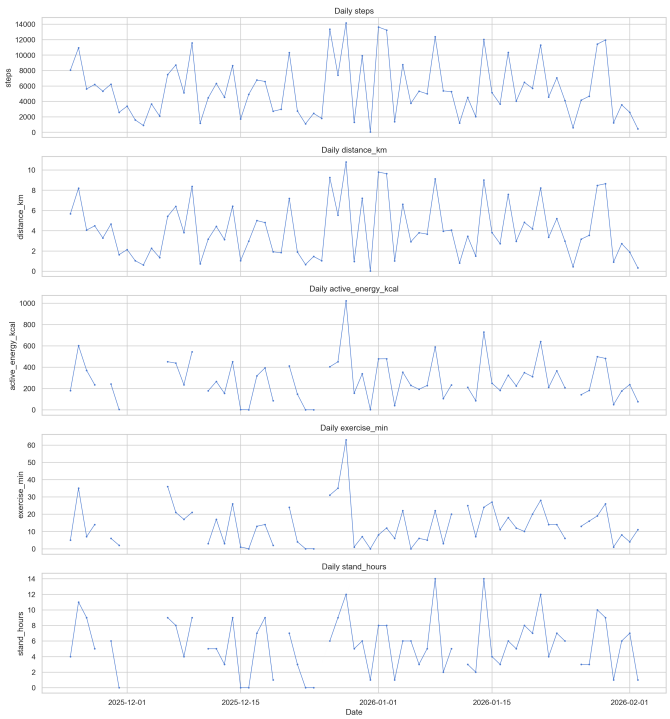

Fig. 1: Time-series plots of core daily metrics (steps, distance, active energy, exercise minutes, and stand hours).

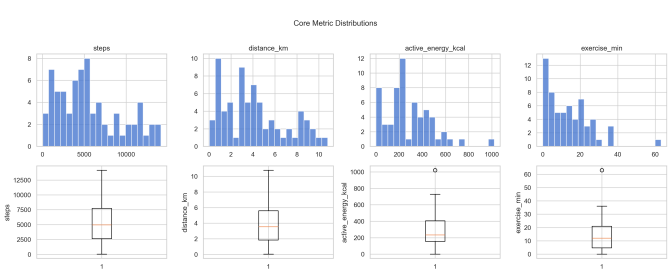

Fig. 2: Distributions (histograms and boxplots) for steps, distance, active energy, and exercise minutes.

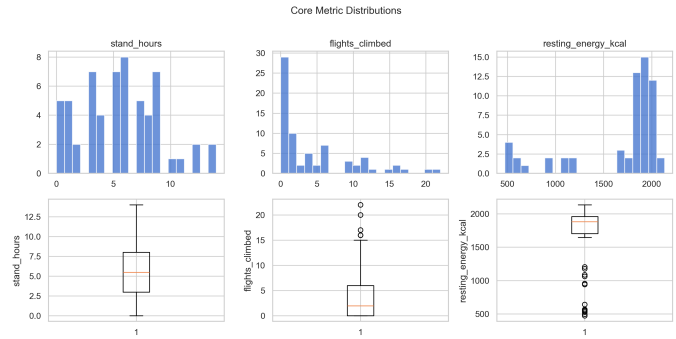

Fig. 3: Distributions (histograms and boxplots) for stand hours, flights climbed, and resting energy.

B. Relationship Structure (Scatter and Correlation)

Fig. 4 visualizes the relationship between steps and active energy.

Pearson correlation between steps and active energy (non-missing rows) was:

$$r = 0.8733, \quad p = 9.16 \times 10^{-20}, \quad n = 60.$$

Steps and distance were nearly collinear (from the heatmap):

$$r \approx 0.9977 \quad (n = 71),$$

which is expected in a within-person dataset because daily distance is largely a linear function of step count given relatively stable step length; heatmap labels are displayed to two decimals.

The correlation heatmap (Fig. 5) further shows that Active Energy is strongly correlated with stand hours (approximately 0.93), distance (approximately 0.89), steps (approximately 0.87), exercise minutes (approximately 0.81), and heart-rate sample count (approximately 0.80), based on the heatmap's 2-decimal display.

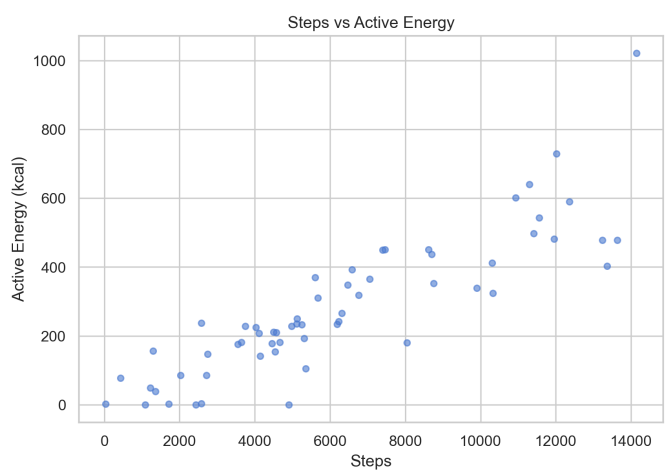

Fig. 4: Scatter plot of steps vs. active energy showing a strong positive association.

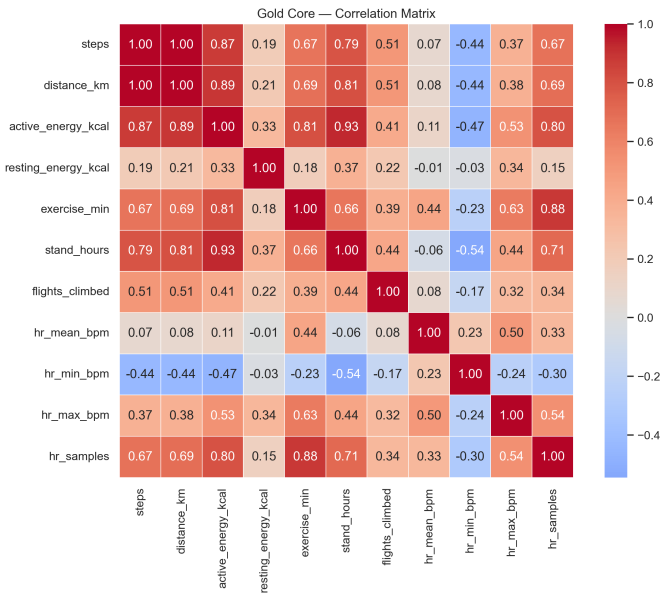

Fig. 5: Gold Core Pearson correlation matrix across daily activity, energy, and heart-rate summary variables.

C. Missingness (Gold Core and Gold Extended)

Gold Core missingness over the 71-day spine was concentrated in heart-rate aggregates and some activity/energy fields. Heart-rate summary variables (hr_mean_bpm, hr_min_bpm, hr_max_bpm) had 23.94% missingness. Active Energy, Resting Energy, Exercise Minutes, and Stand Hours each had 15.49% missingness. Steps, distance, flights climbed, and hr_samples had 0.00% missingness.

Fig. 6 reports missingness for extended physiology variables (Gold Extended additional columns). Respiratory rate was most incomplete (70.42% missing), followed by resting heart rate (39.44%) and HRV (29.58%). Walking speed and step length had 1.41% missingness. Due to high missingness, extended physiology features were excluded from the primary modeling feature set.

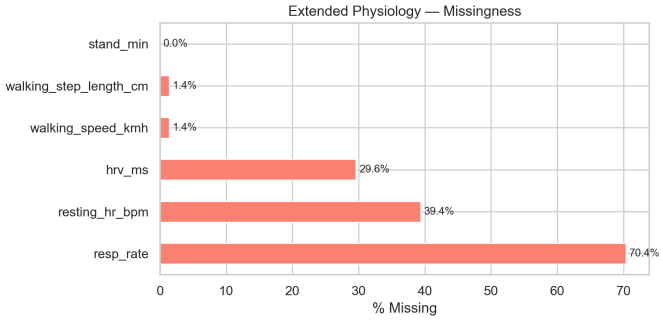

Fig. 6: Percent missingness for extended physiology variables (Gold Extended additional columns).

D. Descriptive Statistical Checks

Normality checks (Shapiro–Wilk) indicated non-normality for both steps and Active Energy:

$$W_{\text{steps}} = 0.9502,\ p = 0.0160; \qquad W_{\text{active_energy}} = 0.9360,\ p = 0.0036.$$

A rank-based robustness check yielded Spearman correlation between steps and active energy:

$$\rho = 0.8961, \quad p = 4.04 \times 10^{-22}, \quad n = 60.$$

Welch's t-test comparing weekday vs. weekend active energy produced: weekday n=43 (mean 267.6 kcal), weekend n=17 (mean 318.4 kcal), t=-0.813, p=0.4239, and Cohen's d=-0.252 (small).

E. Sanity Check: Meters-per-Step

As a plausibility screen for step/distance aggregation errors, a meters-per-step distribution was computed. Fig. 7 shows values concentrated inside a typical human range (approximately 0.3–1.5 m/step), and no days were removed based on this check.

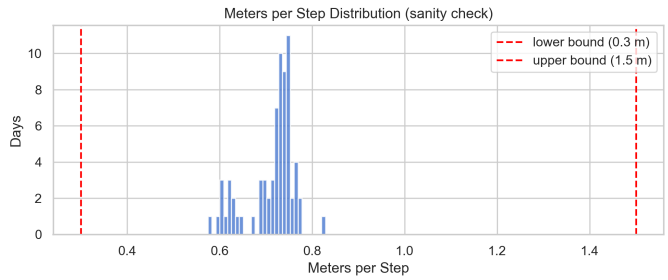

Fig. 7: Meters-per-step distribution used as a plausibility check for distance and step aggregation.

F. Model Performance on the Held-Out Test Set

Supervised learning was performed on a complete-case subset of 46 days, split chronologically into 36 training and 10 test days. Table III reports test-set performance for two baselines and two regression models. Random Forest achieved MAE = 29.19 kcal and R^2 = 0.9448, outperforming Linear Regression (MAE = 50.37 kcal, R^2 = 0.8619). The baseline predictors performed substantially worse on this held-out segment.

TABLE III: Held-Out Test Set Performance (chronological 80/20 split; test n = 10 days)

Model	**MAE (kcal)**	**R^2**
Random Forest Regressor	29.19	0.9448
Linear Regression (OLS)	50.37	0.8619
Dummy (mean)	153.99	-0.3381
Naive (yesterday)	188.39	-1.6713

Fig. 8 visualizes predicted vs. actual values for each model on the test set with date labels. Fig. 9 shows predicted-vs-actual and residual diagnostics. Random Forest predictions more closely track the identity line, consistent with the lower MAE and higher R^2.

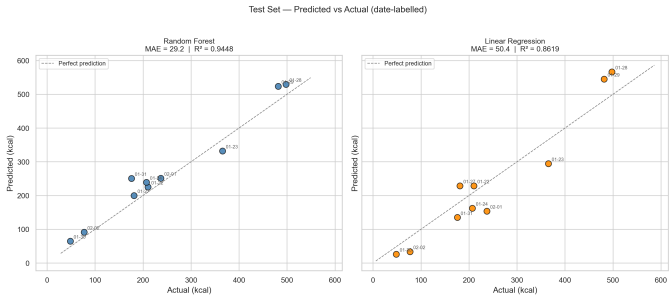

Fig. 8: Test set predicted vs. actual active energy (date-labelled) for Random Forest and Linear Regression.

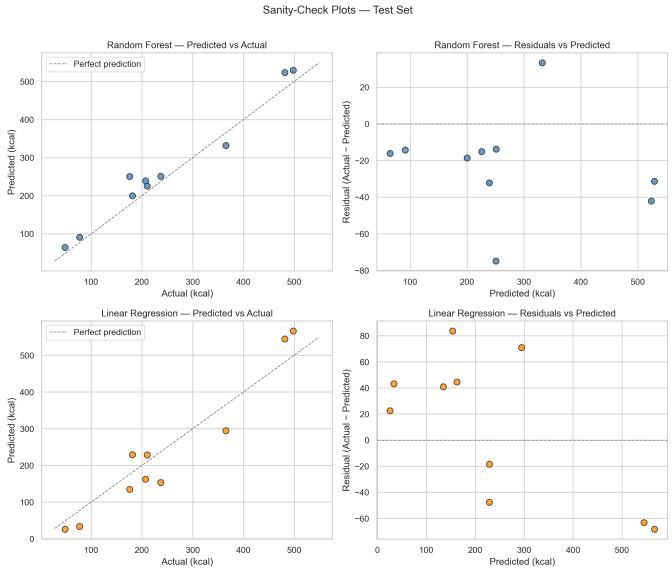

Fig. 9: Sanity-check plots on the test set: predicted vs. actual and residuals vs. predicted for both models.

G. Walk-Forward Time-Series Cross-Validation Summary

Walk-forward time-series cross-validation using TimeSeriesSplit with 5 folds (n_splits = 5) produced lower and more variable performance than the single held-out test evaluation. Table IV summarizes mean and standard deviation across folds.

TABLE IV: Walk-forward time-series cross-validation summary (5 folds; mean ± SD)

Model	MAE (kcal)	R^2
Random Forest Regressor	60.63 ± 48.89	0.8396 ± 0.1810
Linear Regression (OLS)	104.65 ± 58.21	0.5918 ± 0.3699
Dummy (mean)	176.02 ± 50.91	-0.0806 ± 0.0985

H. Random Forest Feature Importance

Fig. 10 reports the top feature importances from the trained Random Forest. Same-day movement features dominate (stand hours, steps, distance), while lag-1 and day-of-week features contribute comparatively less, consistent with a same-day explanatory framing.

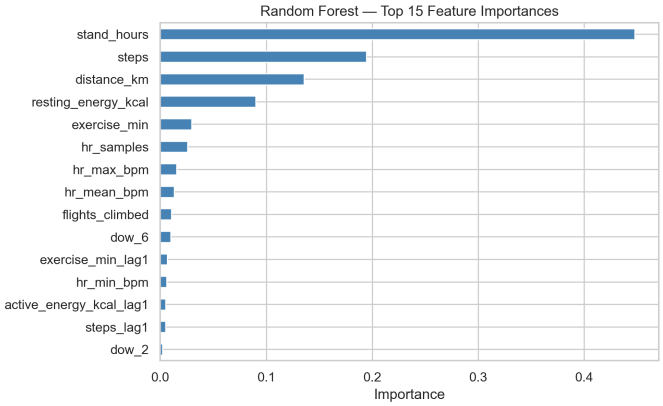

Fig. 10: Random Forest top-15 feature importances for predicting daily active energy.

V. Discussion

This study examined within-person, day-level relationships between Apple Health daily summaries and Apple Health Active Energy (kcal). The results support a strong same-day association between movement-derived metrics and Apple Health's active energy estimate, and show that multivariate regression can predict daily active energy with low test error in this single-subject dataset.

A. Interpretation of Results

EDA indicates that active energy is tightly coupled to daily movement. Pearson and Spearman correlations between steps and active energy were strong and positive:

$$r = 0.8733; \qquad \rho = 0.8961 \quad (n = 60 \text{ non-missing rows}).$$

Shapiro–Wilk tests suggested non-normality for both variables, supporting inclusion of rank-based correlation as a robustness check.

Modeling results further reinforce this relationship. Random Forest produced the best held-out performance (MAE = 29.19 kcal; R^2 = 0.9448), and feature importances emphasized same-day movement variables (especially stand hours, steps, and distance). Linear Regression performed worse, which is consistent with (1) nonlinearities and interactions in the mapping from activity summaries to the Active Energy estimate, and (2) multicollinearity among predictors such as steps and distance (near-collinear in this dataset).

Walk-forward time-series cross-validation produced lower and more variable scores (Random Forest mean R^2 = 0.8396; SD = 0.1810), which is expected with small samples because early folds train on few days and behavior can be non-stationary. This supports the use of time-aware evaluation rather than random shuffling.

B. Comparison to Related Work

Prior work describes Apple Watch as a widely used wearable relying on motion and physiological sensors to estimate daily activity and energy outcomes. The strong same-day alignment

found here is consistent with descriptions of Apple Watch activity rings and energy estimation pipelines, and with the broader literature indicating useful estimates in free-living settings while retaining context-dependent error.

This study does not validate true energy expenditure against indirect calorimetry. Instead, it models Apple Health's *own* active energy estimate using other Apple Health summaries. Therefore, strong predictive performance should be interpreted as internal consistency among Apple Health-derived measures rather than external validation of energy expenditure.

C. Limitations

Several limitations affect interpretation:

- **Single subject (n = 1) and short window.** Findings characterize within-person relationships over 71 days and do not generalize to a population.
- **Complete-case modeling (46/71 days).** Dropping missing days may bias results toward "good recording" days and underrepresent non-wear or irregular behavior.
- **Temporal dependence.** Daily observations are serially correlated; inferential p-values should be treated as descriptive.
- **Construct overlap.** Active Energy is an Apple Health algorithmic estimate derived from related signals; high same-day R^2 reflects alignment with Apple Health's estimate rather than validation against ground truth.
- **Extended physiology missingness.** Several physiology signals (e.g., respiratory rate, HRV) had high missingness and were excluded from primary modeling.

D. Recommendations and Future Work

Future work can strengthen both inference and usefulness:

- Extend the time horizon to increase sample size and capture more behavioral regimes.
- Add contextual covariates (sleep, illness markers, schedule, weather) to explain residual variation.
- Evaluate additional time-series approaches (e.g., regularized regression, boosted trees, autoregressive models) and compare models with and without lagged target features.
- If feasible, compare Apple Health Active Energy to an external reference to move from internal modeling to validation.

VI. Conclusion

This project built reproducible daily datasets from Apple Health exports and modeled Apple Health Active Energy (kcal) using same-day activity summaries in a single-subject setting with time-aware evaluation. Using 71 consecutive days (2025-11-24 to 2026-02-02), we constructed "Gold Core" and "Gold Extended" daily datasets with explicit quality checks, including conversion of 11 suspicious zero-energy days to missing values and a meters-per-step plausibility screen.

Results showed that same-day movement-derived variables (stand hours, steps, distance, and exercise minutes) were strongly and positively associated with Active Energy. On a chronological held-out test set, Random Forest outperformed Linear Regression (MAE = 29.19 kcal, R^2 = 0.9448). Walk-forward cross-validation produced lower and more variable performance (Random Forest mean R^2 = 0.8396, SD = 0.1810), consistent with small, temporally dependent within-person data and reinforcing the need for time-aware validation [5].

Overall, Apple Health daily activity summaries explain Apple Health's same-day Active Energy estimate well in a single-person setting. However, these strong predictions should be interpreted primarily as internal consistency within Apple Health data rather than external validation of true energy expenditure, since Active Energy itself is an algorithmic estimate derived from overlapping signals and wearable energy-expenditure estimates can exhibit substantial error relative to indirect calorimetry [6]. Key limitations include non-random missingness/non-wear, temporal dependence, and construct overlap between predictors and the target.

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